

Soham Roy

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EDUCATION

University of Engineering and Management, Jaipur

B.Tech in Computer Science Engineering

Class 12: GPA 8.64

2024 – Present

CGPA: 9.14/10 (S1: 9.38 — S2: 8.73 — S3: 9.30)

Class 10: GPA 9.89

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C

Frontend: React.js, Tailwind CSS, Vite, Recharts, Leaflet.js

Databases: PostgreSQL, MongoDB, SQLite

ML & Data Science: Scikit-learn, LightGBM, XGBoost, Pandas, NumPy, Matplotlib, Feature Engineering

Domain Knowledge: Quantitative Finance, Geospatial Analysis, IoT Systems

Backend: FastAPI, Node.js, Express.js

Tools: Git, Arduino, ESP32, WordPress

PROJECTS

Stellar Analytics | *FastAPI, Node.js, Scikit-learn, LightGBM, React* | [GitHub](#) | **Winner – IIT BHU 2026** 2026

- Built an end-to-end exoplanet ML pipeline on NASA Kepler data (9,564 records, 50 features), benchmarking 16 models; best classifier achieved ROC-AUC 0.9504 and best regressor R^2 0.9635, MAE 0.14 Earth radii.
- Architected a three-tier system (FastAPI → Express → React) with multi-layer input validation, stratified 80/20 splits, and 15 interactive science visualizations across the full ML pipeline.

AlgoZen – Quant Trading System | *Python, Pandas, NumPy, Custom Backtesting Engine* | [GitHub](#) 2026

- Built a regime-aware NIFTY 500 equity strategy with momentum, RSI, volatility, and trend signals; walk-forward backtesting achieved 10.92% CAGR, Sharpe 1.129, max drawdown 13.05%, outperforming benchmark by 4.07%.
- Engineered risk management with volatility targeting, ATR trailing stops, and circuit breakers; delivered 7.82% alpha, 61.3% monthly win rate, and 175.7% total return (Beta 0.453).

Carbon-Wise | *React, Node.js, Express, SQLite* | [GitHub](#) 2025

- Built a full-stack vehicle lifecycle carbon platform covering manufacturing, well-to-wheel, and disposal emissions; integrated India's state-level grid intensity data to model EV emissions across all 28 states.
- Implemented a 3-vehicle comparison engine, usage-aware recommendation system, and greenwashing detection module via a RESTful Express/SQLite backend with 8 API endpoints.

ShadowMap – Urban Heat Island Prediction | *FastAPI, React, Leaflet, XGBoost* | [GitHub](#) 2025

- Built a geospatial ML system predicting surface temperature across 300 Delhi blocks using OSM footprints, Sentinel-3 LST, and Landsat NDVI; XGBRegressor achieved R^2 0.926, MAE 0.88°C, RMSE 1.13°C with spatial CV R^2 0.906.
- Deployed quantile regression for 80% prediction intervals ($\pm 0.7^\circ\text{C}$) and a what-if simulator for tree cover, roof albedo, and building density interventions on a block-level Leaflet choropleth map.

Smart Road Condition Monitoring System (Funded by IEM Grant Program) | *Arduino, ESP32, IoT Sensors* 2024

- Designed an IoT sensor array system for real-time pothole detection, road quality grading, and accident alerts; engineered SOS integration compatible with navigation system APIs.

EXPERIENCE & LEADERSHIP

Core Organizer – ACEHack 5.0 2026

Rajasthan's Premier Student-Run Hackathon

- Led end-to-end operations for one of Rajasthan's largest student hackathons with 10,000+ registrations, coordinating teams, sponsors, and logistics for 500+ on-site participants.

Web Developer – International Academic Conferences 2025 – 2026

UEM Jaipur

- Designed and deployed production websites for two international conferences – [ICONIC](#) and [ICCASA 2026](#) – serving a global academic audience; built custom HTML/CSS/JS frontends alongside WordPress CMS, owning content, UI, and performance end-to-end.

ACHIEVEMENTS

- Winner** – Stellar Analytics Competition, Technex '26, IIT BHU
- Top 3** – Case Study Competition, Technex '25, IIT BHU
- Round 5 Qualifier** – National Hackathon, MNIT Jaipur
- Top 5 Finalist** – HackUEM